

Approval: **03/03/2025**

Expiration: **03/31/2028**

[BUA Setup](#) ✓

Document Name : BUA-R233

Group: [Instructional Support - BIOE](#) BIO

Rooms: Stanley Hall - B144, B146

Principal Investigator Details:

Name: Chris Anderson

Email: jcanderson@berkeley.edu

Phone: +1 510-666-3611

Assigned Biosafety Admin(s): Olga Draper

Assigned IBC Member(s): Stephen Brohawn, Roberto Zoncu

[Personnel](#) ✓

Chris Anderson Principal Investigator

What type of role does this person have on the project?

☒ PI

This person will be working with the following:

☒ Recombinant or synthetic nucleic acids

Gerard Marriott Member



What type of role does this person have on the project?

- ☒ PI
- ☒ Other

Describe this person's responsibilities:

Instructor of BioE 163L

This person will be working with the following:

- ☒ Others

Specify materials:

None of the above.

Mary West Member

What type of role does this person have on the project?

- ☒ Other

Describe this person's responsibilities:

Lab manager is close but Dr. West runs the core facility next door and helps keep an eye on the room, users and equipment.

This person will be working with the following:

- ☒ Others

Specify materials:

None of the above.

Locations 

 Stanley Hall—B144

Is this a shared space?

Yes

Space usage:

- ☒ Laboratory

If this space has other usages, specify them here:

Instructional

Space containment level (biosafety level):

BSL-1

Use this textbox to add field sites or non-UC Berkeley sites where biohazardous materials are used or stored:

—

 Stanley Hall—B146

Is this a shared space?

Yes



Space usage:☒ Laboratory**If this space has other usages, specify them here:**

There are 4 biosafety cabinets, currently and planned use at BSL1.

Space containment level (biosafety level):

BSL-1

Use this textbox to add field sites or non-UC Berkeley sites where biohazardous materials are used or stored:

—

Project Overview ✓**Project Title:**

Instruction of undergraduate and graduate students in laboratory techniques and theory.

Provide a brief summary of your research goals and specific objectives: It is important to use language that is general enough to be understood by people with non-scientific backgrounds and to define acronyms the first time they are used.

The instructional program includes six undergraduate and four graduate courses serving over 200 students per academic year, plus summer bootcamps that train an additional 100 students. These offerings support a continuum of training in bioengineering from first-time wet lab exposure to advanced undergraduate research.

Fall Semester:

* BioE 140L Synthetic Biology Lab: Provides hands-on experience in genetic circuit design and molecular cloning in E. coli for upper-division undergraduates.

* FERB (Foundations of Engineering Biology II DeCal): A course for first- and second-year undergraduates that introduces cloning, transformation, and microbial assays in E. coli.

* iGEM (International Genetically Engineered Machine): A year-long synthetic biology program where students engineer novel systems in E. coli and Lactococcus lactis. Projects include biosynthesis, CRISPR tools, and antifungal proteins.

* BioE 98/198 DeCal: Explorations in Bioprinting (also offered Spring): A student-led course focused on 3D biomaterials printing, with possible use of rodent cell lines.

Spring Semester:

* BioE 115/215 Tissue Engineering Lab: Enrolls approximately 60 students in mammalian tissue culture using rodent cell lines.

* BioE 163L/263L Optical Biosensors Lab: Covers design and optimization of fluorescence-based biosensors; does not typically involve biohazards.

* DeCal and iGEM continue through Spring.

Summer:

* QB3 Bootcamp: Eight one-week sessions training approximately 18 students per session in molecular biology basics using E. coli K-12 and standard plasmids.

All activities described here are conducted using RG1 organisms under BSL1 conditions and standard recombinant DNA procedures.

Biological Materials ✓

Select the biological organisms and materials to be used: organisms, cells, tissues, body fluids, viral vectors, toxins, and genome modifiers (found under "Cas9"). For other genes and plasmids, please fill out the [Recombinant DNA Template](#) and upload it in the **Attachments section.**

- Escherichia coli (non-pathogenic) - Type: Bacteria

Biological Safety Level (BSL):

BSL-1

Strain/type:

E. coli K12 and derivatives

This microbial agent is pathogenic to:

☒ Not applicable (e.g., E. coli BL21)

Source of material:

☒ Domestic - Commercial

Is a permit required to obtain or possess this organism?

No

Will this organism be used or put in storage only?

Used

This organism will be added to the following tissues or cell lines:

☒ Not applicable

This organism will be added to the following whole organisms:

☒ Not applicable

Select the procedures that apply:

☒ Centrifugation, Ultra-centrifugation

☒ Pipetting

☒ Shaking culture

☒ Streak plating

☒ Vortexing

Will this organism be, or is it already, genetically modified?

Yes

Will greater than 10 liters (per container) of this material be cultured at any given time?

No

Will/Does this modification alter the organism's pathogenicity?

No

- Animal Mouse cells - Type: Animal Primary Cells

Source organism:

Vertebrate

Vertebrate name:

Mus musculus -- ESTABLISHED CELL LINES

☒ Cells

Primary or established cells:

Established cell lines

Provider/source of biological materials:

on campus cell culture core, commercial sources, neighboring labs

Is this cell line known to contain pathogens?

No



Will this material be used or put in storage only?

Used

Select the procedures that apply:

- ☒ Centrifugation, Ultra-centrifugation
- ☒ Pipetting

Will this material be genetically modified?

No

Will this material be used to produce a viral vector?

No

This material will be added to the following whole organisms:

- ☒ Not applicable

- Lactococcus lactis - Type: Bacteria

Biological Safety Level (BSL):

BSL-1

Strain/type:

NZ9000

This microbial agent is pathogenic to:

- ☒ Not applicable (e.g., E. coli BL21)

Source of material:

- ☒ Domestic - Collaborator/Individual

Is a permit required to obtain or possess this organism?

No

Will this organism be used or put in storage only?

Used

This organism will be added to the following tissues or cell lines:

- ☒ Not applicable

This organism will be added to the following whole organisms:

- ☒ Not applicable

Select the procedures that apply:

- ☒ Centrifugation, Ultra-centrifugation
- ☒ Electroporation
- ☒ Pipetting
- ☒ Shaking culture
- ☒ Streak plating
- ☒ Vortexing

Will this organism be, or is it already, genetically modified?



Yes

Will greater than 10 liters (per container) of this material be cultured at any given time?

No

Will/Does this modification alter the organism's pathogenicity?

No

NIH Guidelines

National Institutes of Health (NIH) sections that apply to this project:

- ☐ III-A (Experiments that Require Institutional Biosafety Committee Approval, RAC Review, and NIH Director Approval Before Initiation)
- ☐ III-B (Experiments That Require NIH OSP and Institutional Biosafety Committee Approval Before Initiation)
- ☐ III-C (Experiments that Require Institutional Biosafety Committee and Institutional Review Board Approvals and RAC Review (if applicable) Before Research Participant Enrollment)
- ☒ III-D (Experiments that Require Institutional Biosafety Committee Approval Before Initiation)
 - ☐ Section III-D-1. Experiments Using Risk Group 2, Risk Group 3, Risk Group 4, or Restricted Agents as Host-Vector Systems
 - ☒ Section III-D-2. Experiments in Which DNA From Risk Group 2, Risk Group 3, Risk Group 4, or Restricted Agents is Cloned into Nonpathogenic Prokaryotic or Lower Eukaryotic Host-Vector Systems
 - ☐ Section III-D-3. Experiments Involving the Use of Infectious DNA or RNA Viruses or Defective DNA or RNA Viruses in the Presence of Helper Virus in Tissue Culture Systems
 - ☐ Section III-D-4. Experiments Involving Whole Animals
 - ☐ Section III-D-5. Experiments Involving Whole Plants
 - ☐ Section III-D-6. Experiments Involving More than 10 Liters of Culture
 - ☐ Section III-D-7. Experiments Involving Influenza Viruses
- ☒ III-E (Experiments that Require Institutional Biosafety Committee Notice Simultaneous with Initiation)
 - ☐ Section III-E-1. Experiments Involving the Formation of Recombinant or Synthetic Nucleic Acid Molecules Containing No More than Two-Thirds of the Genome of any Eukaryotic Virus
 - ☐ Section III-E-2. Experiments Involving Whole Plants
 - ☐ Section III-E-3. Experiments Involving Transgenic Rodents (the animal's genome has been altered by stable introduction of recombinant or synthetic nucleic acid molecules; only experiments that require BSL-1 are covered by this section, for higher containment levels, see III-D-4)
- ☒ III-F (Exempt Experiments)

Select all the checkboxes that apply:

 - ☒ Section III-F-8. Those that do not present a significant risk to health or the environment (see Section IV-C-1-b-(1)-(c), Major Actions), as determined by the NIH Director, with the advice of the RAC, and following appropriate notice and opportunity for public comment. See Appendix C, Exemptions under Section III-F-8 for other classes of experiments which are exempt from the NIH Guidelines.

Other Sections or Appendices:

-
- ☐ I need help with this Section.
 - ☐ None apply.



Experimental Procedures and Risk Assessments ✓

- Please describe the experimental procedures to be performed, including the use of all materials listed in this BUA.
- Indicate the relationship between materials (e.g. which genes will be carried by which viral vector for transduction of which cells/host; or which organism/cells will be infected with which microorganism).
- Please create separate sections based on biosafety levels (e.g. BSL1, ABSL1, BSL2, etc.) and for each BSL section, describe the spill, disinfection, and waste management protocols as well as the PPE to be worn.
- Please visit our [Attachments page](#) to download descriptions of common biosafety practices. You can edit and paste the relevant text into the text box below.

The following instructional and project-based lab activities are conducted under BSL1 conditions, using RG1 organisms and standard recombinant DNA procedures. No RG2+ materials, toxins, or pathogens are used:

* QB3 Bootcamp (Summer): Students practice basic molecular biology techniques using E. coli K12 and standard plasmids. Activities include PCR, restriction digest, ligation, bacterial transformation, colony screening, and plasmid purification. Students clone GFP into pUC-based plasmids and confirm their constructs via sequencing.

* BioE 98/198 DeCal: Explorations in Bioprinting: Students learn to 3D print biomaterials. The course may use unmodified rodent cell lines, though it typically focuses on physical biomaterials rather than recombinant DNA techniques.

* BioE 115/215 Tissue Engineering Lab: Students work with unmodified rodent cell lines to learn mammalian tissue culture techniques. No recombinant DNA or microbial work is performed.

* FERB (Foundations of Engineering Biology II DeCal), BioE 140L, and iGEM: These programs involve more advanced recombinant work.

- Pimaradiene Bioproduction (BioE140L):

E. coli W and Mach1 are engineered to biosynthesize pimaradiene, isopimaradiene, and lycopene through the terpenoid pathway. Constructs include plasmids with terpene cyclases and supporting enzymes.

- Engineered Cell Lysis (iGEM):

Self-lysis systems using holins, lysozymes, spanins, and antiholins from RG1 organisms are cloned into plasmids or integrated into the genome of E. coli K12, W, Mach1, or BL21. Promoters include Pbad, Ptet, and Plac. Libraries with altered regulatory sequences or signal peptides may be generated.

- Antifungal Biopreservatives (iGEM):

Chitinases and antifungal proteins from RG1 organisms (e.g., L. lactis, B. cereus, S. marcescens) are expressed in E. coli and L. lactis. Genes are cloned into E. coli/L. lactis shuttle vectors and tested for antifungal activity against Penicillium commune. Assays include growth inhibition and protein quantification. Feta cheese may be produced under lab conditions for functional testing but will not be consumed.

- Engineering Synthetic Promoters (iGEM, FERB, BioE140L):

Libraries of synthetic promoters and terminators are constructed in E. coli using randomized sequences surrounding core promoter elements. Fluorescent reporters such as amilGFP are used to characterize expression. Promoters may include regulatory elements responsive to nutrients, temperature, or ions.

- CRISPR Editing Tools (iGEM):

Novel CRISPR-based genome editing systems are developed in E. coli K12, W, Mach1, or BL21 using Cas9, Cas12a, and LscB along with RG1 exonucleases and ligases. Constructs may include R6K or F plasmid origins. Selection assays include antibiotic resistance, origin disruption, and loss-of-function in sensitivity genes. Proteins may be purified for in vitro study.

* General Laboratory Practices:

All lab members must wear full-length leg coverings, closed-toe shoes, and appropriate personal protective equipment (PPE), including lab coats, gloves, and eye or face protection as needed. No eating,

drinking, or makeup application is permitted in the lab. Hands must be washed before exiting. Guests must be escorted by trained lab personnel at all times.

*** Sharps Safety:**

Sharps safety-engineered devices, plastic blades, and alternatives such as plastic Pasteur pipettes are used whenever possible. Contaminated needles are not recapped, bent, or removed from syringes. All contaminated sharps are disposed of in FDA-approved, puncture-resistant, and leak-proof sharps containers, sealed when <75% full. Reusable sharps are cleaned and stored in sealed hard plastic containers or their protective covers.

*** Bacterial Growth and Culture Handling:**

Bacteria including *E. coli* and *L. lactis* will be cultivated using LB, M9, or 2YT medium, or appropriate *L. lactis* media in petri dishes, 24-well blocks, test tubes, and shake flasks. Culture volumes will not exceed 10L.

*** Disinfection and Waste Disposal:**

Liquid waste will be autoclaved or treated with 10% bleach for at least 20 minutes before disposal via the drain with copious water.

Solid waste (e.g., petri dishes, pipette tips, and plastics contaminated with biological material or DNA) will be collected in autoclavable bags, sterilized by autoclave or pressure cooker, and disposed of as general waste.

*** Spill Response:**

Spills will be reported to the Instructor and supervisor immediately and to EH&S within 8 working hours (bso@berkeley.edu). The spill area should be cleared of bystanders. PPE must be donned before cleanup. Cover spills with absorbent material and apply 10% bleach starting at the perimeter. Maintain contact time for at least 20 minutes. Clean up with fresh disinfectant-soaked towels and dispose of all materials as solid waste. Broken sharps should be picked up using tongs or a dustpan and disposed of in a sharps waste container. Hands must be washed after cleanup.

*** Injury and Exposure:**

EH&S must be contacted immediately in the case of injury or exposure: 510-642-3073 or (bso@berkeley.edu). During business hours, seek care at Tang Urgent Care; after hours, go to Alta Bates Hospital. Call 911 (or 510-642-3333 from a cell on campus) for emergencies.

*** Transport:**

All biological materials are transported in sealed, leak-proof, decontaminable secondary containers labeled with lab name and contact information.

*** Supervision**

Students are classified as Trainees, Independents, or Supervisors. Trainees, typically first-time lab participants enrolled through courses like BioE 140L, BioE 98, or independent research units, receive in-lab orientation and pipette training, sign safety agreements, and are always directly supervised. They are not listed as formal Personnel and are not required to complete EH&S training.

Independents are trained upperclassmen who have completed EH&S requirements, reviewed the BUA, signed the Chemical Hygiene Plan, and are listed on the lab roster. They are permitted independent access to lab spaces for BSL1 work.

Supervisors are experienced Independents who receive additional training, attend supervisor meetings, and sign a Supervisor Agreement. At least one Supervisor is present whenever Trainees are working. Labs are scheduled to reduce crowding and eliminate risk, with no access to hazardous chemicals or sharps, and substitutions such as GelGreen in place of ethidium bromide.

- Provide a risk assessment for the known and suspected biohazards posed to humans, animals, or plants by the materials/infectious agents listed in the BUA, including the disease symptoms, transmission routes and information on laboratory acquired infections. Use the following resource for guidance: <https://www.canada.ca/en/public-health/services/laboratory-biosafety-biosecurity/pathogen-safety-data-sheets-risk-assessment.html>
- If an exotic arthropod vector system or a transgenic organism/microorganism is used, discuss the possible consequences of a release into the local environment.
- For work involving recombinant nucleic acids or genetically modified organisms or microorganisms, discuss the risks to human health and the environment posed by the expression of genes of interest, the vectors, and transformed organisms or microorganisms.

The organisms used in the Teaching Lab include rodent cell lines which are non-transgenic and pose no risk to the environment, and E. coli K12 strains of bacteria that are non-pathogenic and pose little risk to humans or the environment.

The recombinant nucleic acids used in lab are standard plasmids such as pUC derivatives containing a bacterial ampicillin resistance gene to determine successful transformation and efficiency. These have been shown over the past many decades to have little risk to humans or the environment. However, due to their recombinant nature, all work will be done at BSL1.

Risk Minimization ✓

Will access to the laboratory (or facility) be restricted while work is in progress?

Yes

Will laboratory (or facility)-specific safety training be provided to personnel?

Yes

Will nucleic acids be used in this research project?

Yes

Select the appropriate Personal Protective Equipment (PPE) required for this type of work:

- ☒ Regular lab coat
- ☒ Gloves
- ☒ Safety glasses

Will bloodborne pathogens be used in this research project?

No

Will infectious agents be used in this research project?

No

Will sharps be used in this research project?

Yes

Sharps type

- ☒ needles
- ☒ razors

Which manipulations will involve sharps?

A razor is sometimes used to cut and shape structures made of PDMS, gelatin or similar. A needle is sometimes used to fill a structure with a liquid or polymer. Neither is typically used with biologicals.

Glass slides and coverslips are used to observe cells, and they may break.

Will training for the safe handling of sharps be provided?

Yes

What would happen in the event of auto-inoculation (with the agent or genes of interest)?

If there was an stick or cut incident involving cells or recombinant DNA, it would be treated by washing the area well and applying first aid bandaging, seeking medical attention at the Tang center or nearest urgent care (eg. Alta Bates), and reporting it to EH&S within 8 hours.

Describe any additional sharps safety practices that will be used:

Standard sharps safety practice.

Will a centrifuge be used in this research project?

Yes

What biohazardous materials will be centrifuged?

Rodent and bacterial cells will be sedimented by centrifugation.

Nucleic acids (plasmids) will be processed using a microfuge as per mini-prep protocols.

Select the safety practices related to centrifugation that will be followed:



Personnel will be trained in the safe use of centrifuges

Describe any additional centrifugation safety practices that will be used:

—

Will biological safety cabinets be used in this project?

Yes

Location(s):

B146

Describe procedures that will be performed in the biological safety cabinet(s):

Sterile culture of rodent cells.

Will any activity be performed that might generate aerosols, which present a human hazard, other than centrifugation?

Yes

Select the associated aerosol generating procedures:



Materials will be pipetted



Shaking culture



Vortexing

Select safety practices followed for any aerosols generating procedures that can present a human hazard:

Personnel will be trained in proper pipetting techniques:

Yes

Will animals be used in this research project?

No

Will plants be used in this research project?

No

Describe any additional risk minimization safety practices:

—

Waste Disposal ✓

Indicate your methods for terminal inactivation and waste disposal of the biological materials. If an autoclave will be used to inactivate waste (liquid or solid), the autoclave must be tested with a biological indicator at least monthly.

Will liquid waste be generated?



Yes

Liquid waste will be disposed of in the following way(s):

- ☒ Inactivation with bleach (10% final concentration, with 30 minutes of contact time, and then drain disposed).

Indicate the contaminants in the waste:

Rodent cells and any microbial contaminants in the culture of such cells.

- ☒ An autoclave will be used (121°C, 15 psi, 30 minutes), and then drain disposed.

Indicate the contaminants in the waste:

Bacterial cell culture.

- ☐ Inactivation in vacuum traps
- ☐ Mixed waste, e.g. formalin fixed cells. (will be disposed through HWP)

Will solid waste be generated?

Yes

Solid waste will be disposed of in the following way(s):

- ☒ Autoclave (121°C, 15 psi, 30 minutes) in clear autoclave bags, using an indicator (autoclave tape or, preferably, a steam indicator strip).
- ☒ Autoclaved BSL1 waste is then disposed of in the regular trash.

Indicate the contaminants in the waste:

Bacterial plates, cell culture plates, dishes, flasks and serological pipets.

- ☐ Solid BSL2 waste is disposed of as red bag waste via Stericycle.
- ☐ Soil - Autoclaved for 60 minutes (121°C, 15 psi) in pans or clear autoclave bags, using an indicator (autoclave tape or, preferably, a steam indicator strip).
- ☐ Mixed waste. Any mixed waste (chemicals and biologicals) should be disposed of through HWP.

Will sharps waste be generated?

Yes

Sharps will be disposed of in the following way(s):

- ☒ Medical sharps waste: sharps containers with the biohazard symbol and the word Biohazard will be used. (Sharps containers will be closed when 2/3 full and transported to the medical waste accumulation site within thirty days of reaching the fill line.) Add a bar code sticker provided by the Hazardous Waste Program.

Indicate the contaminants in the waste:

Most razors or needles should not be exposed to biologicals, but there may be occasional glass sharps from coverslips or slides.

- ☐ Sharps waste with mixed hazard contaminants: label the sharps container with the hazards to be collected (e.g. hazardous chemicals, radiological waste, biohazardous waste, non-biohazardous waste). Close the container when 2/3 full and create a hazardous Waste Program label for pick up within 30 days of reaching the fill line.

Will carcass or tissue (Pathology) waste be generated?

No

If you will be using a terminal inactivation or waste disposal method that is not already described, please describe the method:

—

Attachments ✓Please visit our [Attachments page](#) to download forms you may need.Upload your [recombinant DNA templates, lentiviral exposure control plan, SOPs](<https://ehs.berkeley.edu/safety-subjects/biological-safety/bua-attachments>), and permits here.

[Conboy nee West BUA #233, BSL1 appr 2025.03.pdf](#)

[2025_05_29-recombinant-nucleic-acid-template.docx](#)

Shipping and Transport ✓

The following types of materials will be transported:

☒ Not applicable

BUA Certification ✓

Acknowledgement of Responsibilities:

I will ensure that ALL persons who enter my laboratory or facility will be instructed about the hazards associated with this project and will practice appropriate safety precautions.

I will ensure that ALL persons conducting this work (including my collaborators) have received instruction about the specific hazards associated with the work and the specific safety equipment, practices and behaviors required during the course of the work and use of these facilities. I will ensure that all authorized personnel listed in this BUA (except for administrative staff who will not handle the materials listed on this BUA) will complete all required biosafety training that is related to their work and the materials they will handle. The IBC or EH&S may review my records documenting this requirement and may enter my laboratory or facility at any time to review my operations.

If there is a spill of biohazardous material, or any equipment or facility failure (e.g., ventilation failure) that may result in potential exposure to laboratory personnel, the general public, or loss of containment of biohazardous material, I will report the incident to the EH&S Biosafety Office within 8 hours if a campus laboratory and 4 hours for any external entity or start-up. Should an employee or coworker become ill and exhibit symptoms and signs consistent with an infection by an organism associated with my research, I will report the incident to the EH&S Biosafety Office immediately.

Any changes to the BUA, especially, changes in materials, work, location, or procedures performed are required to submit an amendment to obtain an approval by the IBC or Biosafety Office prior to commencing work. (Examples: 1. Change from mouse cells to working with human or primate cells; 2. Change from adenoviral vector to HIV [lentiviral] vector.) Work with biohazardous materials or recombinant organisms / nucleic acids shall not be conducted until approved by the IBC or the Biosafety Office.

I will ensure that ALL risk mitigation strategies and safety practices specified in this BUA are understood and followed by ALL personnel. I will have all persons working in the space listed in this BUA aware of the hazards associated with this project.

I will comply with all of the applicable sections of the NIH Guidelines for Research Involving Recombinant or Synthetic Nucleic Acid Molecules.

I agree with all of the above

Dual Use Research of Concern:

My research project described in this BUA involves an [agent/toxin listed in section 6.2.1.](#) of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern

No

